Supplementary Figure S12 — Single-gene cis-eQTL × AF-GWAS regime check is conservative at current atrial-appendage panel power; framework-level chamber-reversal genes (Fig. 6c) supply the gene-level positive evidence

Anchor	Axis	cis-eQTL primary GTEx atrial appendage	cis-eQTL sensitivity GTEx whole blood	AF GWAS min p (window)	coloc PP4 (primary)	Verdict
PRELP	matrix	○ lead p = 2.6e-4	○ lead p = 4.3e-4	6.6e-84	0.03	not supported
OMD	matrix	○ lead p = 6.1e-4	— *n/a (gene absent in tissue)*	1.3e-7	0.03	not supported
IDH2	metabolic	◖ lead p = 8.2e-6	○ lead p = 7.1e-4	7.8e-9	0.01	not supported
NDUFA8	metabolic	○ lead p = 3.7e-4	○ lead p = 6.6e-5	1.9e-6	0.04	not supported

cis-eQTL gate ● p < 5e-8 ◖ 5e-8 ≤ p < 1e-5 marginal ○ p ≥ 1e-5 gate fail

Verdict supported inconclusive not supported

Plausibility — not causal. cis-eQTL × AF-GWAS coloc on a pre-registered anchor set; primary tissue = GTEx atrial appendage (n = 372); blood sensitivity = GTEx whole blood (n = 670, eQTL Catalogue substitute for eQTLGen). Two canonical AF GWAS loci (PITX2, KCNN3) are shown below the divider as pre-registered positive controls; both fail the same conservative gate at this panel power, indicating that the framework anchor null reflects the calibration of the cis-eQTL panel rather than absence of any single-gene mediation in principle. Verdict bins fixed in spec §6.7; gate = lead cis-eQTL p < 5e-8 within ±1 Mb of the gene.